

EC2 Instance Launch

aws

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Ask Amazon Q

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Divya Erupaka

EC2

Instances

Launch an instance

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices

Take a walkthrough

Do not show me this message again.

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

Techie-Divya

Add additional tags

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

Summary

Number of instances Info

1

Software Image (AMI)

Canonical, Ubuntu, 26.04, amd6...read more

ami-091138d0f0d41ff90

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

CloudShell

Feedback

Console Mobile App

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EC2

Instances

Launch an instance

Success

Successfully initiated launch of instance (i-01d1ac6d0222cc825)

Launch log

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

< 1 2 3 4 5 6 >

Create billing usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

Learn more

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

Connect an RDS database

Create a new RDS database

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

CloudShell

Feedback

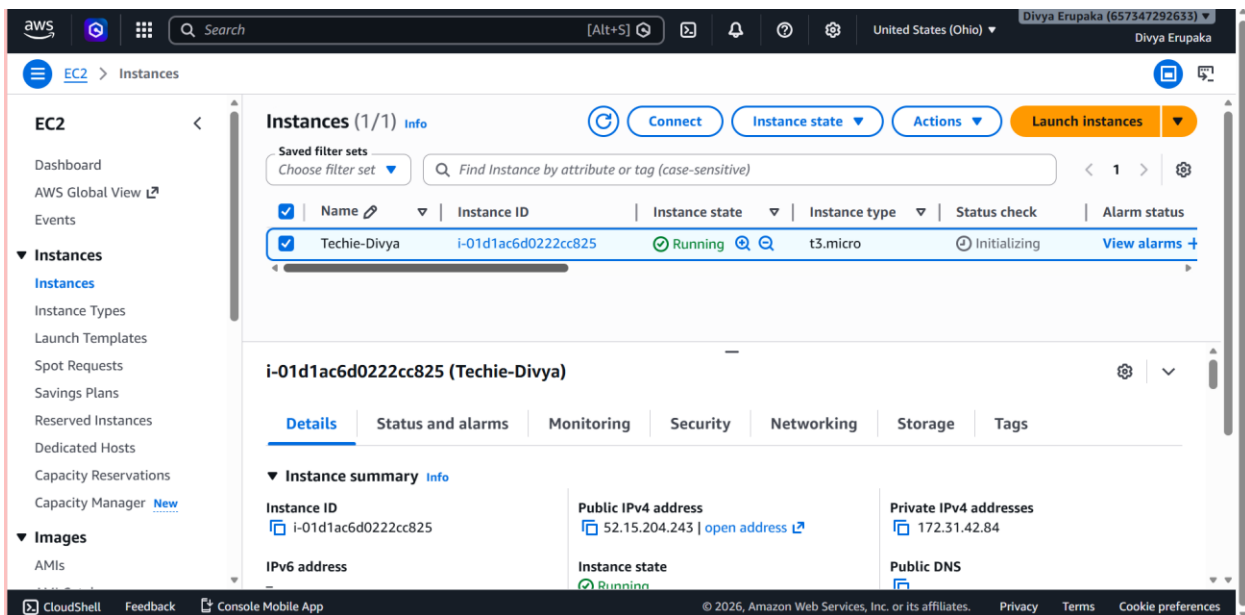
Console Mobile App

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Connected to Instance

```
* Support: https://ubuntu.com/pro

System information as of Thu May 7 11:17:24 UTC 2026

System load: 0.33      Temperature: -273.1 C
Usage of /: 30.4% of 6.61GB    Processes: 121
Memory usage: 28%      Users logged in: 0
Swap usage: 0%         IPv4 address for ens5: 172.31.43.255

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

/usr/bin/xauth: file /home/ubuntu/.Xauthority does not exist
ubuntu@ip-172-31-43-255:~$
ubuntu@ip-172-31-43-255:~$
ubuntu@ip-172-31-43-255:~$
ubuntu@ip-172-31-43-255:~$
ubuntu@ip-172-31-43-255:~$
ubuntu@ip-172-31-43-255:~$
ubuntu@ip-172-31-43-255:~$ sudo apt update
Get:1 http://us-east-2.ec2.archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:2 http://us-east-2.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [136 kB]
```

```
ubuntu@ip-172-31-43-255:~$ sudo apt install openjdk-17-jdk -y
Installing:
  openjdk-17-jdk

Installing dependencies:
  adwaita-icon-theme      libatk-bridge2.0-0t64  libglx0                libsm6                 libxmu6
  alsa-topology-conf      libatk-wrapper-java    libglycin-2-0          libthai-data           libxpm4
  alsa-ucm-conf           libatk-wrapper-java-jni libgraphite2-3         libthai0               libxrandr2
  at-spi2-common          libatk1.0-0t64         libgtk2.0-0t64        libvulkan1             libxrender1
  at-spi2-core            libatspi2.0-0t64       libgtk2.0-bin         libwayland-client0     libxres1
  bubblewrap              libavahi-client3       libgtk2.0-common       libx11-dev             libxshmfence1
  ca-certificates-java    libavahi-common-data   libharfbuzz0b         libx11-xcb1            libxt-dev
  dconf-gsettings-backend libavahi-common3       libheif-plugin-aomdec  libxau-dev             libxt6t64
  dconf-service           libc-dev-bin           libheif-plugin-aomenc  libxaw7                libxtst6
  fontconfig              libc6-dev              libheif1              libxcb-dri3-0          libxv1
  fontconfig-config       libcairo-gobject2      libhwyt64             libxcb-glx0            libxxf86dga1
  fonts-dejavu-core       libcairo2              libibus-1.0-5         libxcb-present0        libxxf86vm1
  fonts-dejavu-extra      libcups2t64            libice-dev            libxcb-randr0          linux-libc-dev
  fonts-dejavu-mono       libdatrie1             libice6               libxcb-render0         luit
  glycin-loaders          libdconf1              libjpeg-turbo8        libxcb-shape0          manpages-dev
  glycin-thumbnails       libdisplay-info3       libjpeg8              libxcb-shm0            mesa-libgallium
  gnome-accessibility-themes libdrm-intel1          libjxl0.11            libxcb-sync1           mesa-vulkan-drivers
  gnome-themes-extra      libfontconfig1         libjms2-2             libxcb-xfixes0         openjdk-17-jdk-headless
  gnome-themes-extra-data libgail-common          libpango-1.0-0        libxcb1-dev            openjdk-17-jre
  gsettings-desktop-schemas libgail18t64           libpangocairo-1.0-0   libxcomposite1         openjdk-17-jre-headless
  gtk-update-icon-cache   libgbm1                libpangoft2-1.0-0     libxcursor1            rpcsvc-proto
  gtk2-engines-pixbuf     libgdk-pixbuf-2.0-0    libpciaccess0         libxdamage1            user-session-migration
  hicolor-icon-theme      libgdk-pixbuf2.0-common libpcsclite1          libxdmcp-dev           uuid-dev
  ibus-gtk                libgif7                libpixman-1-0         libxf86fixes3          x11-common
  java-common             libgl1                 librsvg2-2            libxft2                x11-utils
  libao3                  libgl1-mesa-dri        librsvg2-common       libxi6                 x11proto-dev
  libasound2-data         libglvnd0              libsharpvuv0          libxinerama1           xorg-sgml-doctools
  libasound2t64           libglx-mesa0           libsm-dev             libxkbfile1            xtrans-dev

Suggested packages:
  adwaita-icon-theme-legacy cups-common libheif-plugin-i2kenc libsm-doc fonts-ipafont-gothic
```

```
ubuntu@ip-172-31-43-255:~$ java -version
openjdk version "17.0.18" 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-1, mixed mode, sharing)
ubuntu@ip-172-31-43-255:~$
```

Attempted Jenkins Repository Installation

Initially we tried installing Jenkins using Jenkins repository.

Commands included:

- Adding Jenkins repo
- Importing GPG keys
- Running apt update

Problem faced:

- Ubuntu version was resolute (new Ubuntu release)
- Jenkins repository did not officially support that version yet
- Repository signature verification failed

Error seen:

NO_PUBKEY 7198F4B714ABFC68

Alternative Method — Install Jenkins Using .deb Package

Instead of repository installation, we downloaded Jenkins package manually.

Command used:

```
wget https://pkg.jenkins.io/debian-stable/binary/jenkins_2.504.1_all.deb
```

Purpose:

- Downloads Jenkins Debian package directly from Jenkins official website.

Downloaded file:

- jenkins_2.504.1_all.deb

Install Jenkins Package

Command used:

```
sudo dpkg -i jenkins_2.504.1_all.deb
```

Purpose:

- dpkg installs .deb packages in Ubuntu/Debian systems.
- Jenkins application files and service files were installed.

What happened internally:

- Jenkins user got created
- Jenkins service got registered with systemd
- Jenkins files got placed in:
 - /var/lib/jenkins
 - /etc/systemd/system
 - /usr/share/jenkins

Fix Broken Dependencies (if required)

Command used:

```
sudo apt --fix-broken install -y
```

Purpose:

- Automatically installs missing dependencies required by Jenkins.

Start Jenkins Service

Command used:

```
sudo systemctl start jenkins
```

Purpose:

- Starts Jenkins service immediately.

Internally:

- Jenkins Java process starts running in background.
-

Step 10: Enable Jenkins Service

Command used:

```
sudo systemctl enable jenkins
```

Purpose:

- Enables Jenkins to start automatically whenever server reboots.
-

Step 11: Verify Jenkins Status

Command used:

```
sudo systemctl status jenkins
```

Purpose:

- Checks whether Jenkins service is running properly.

Expected output:

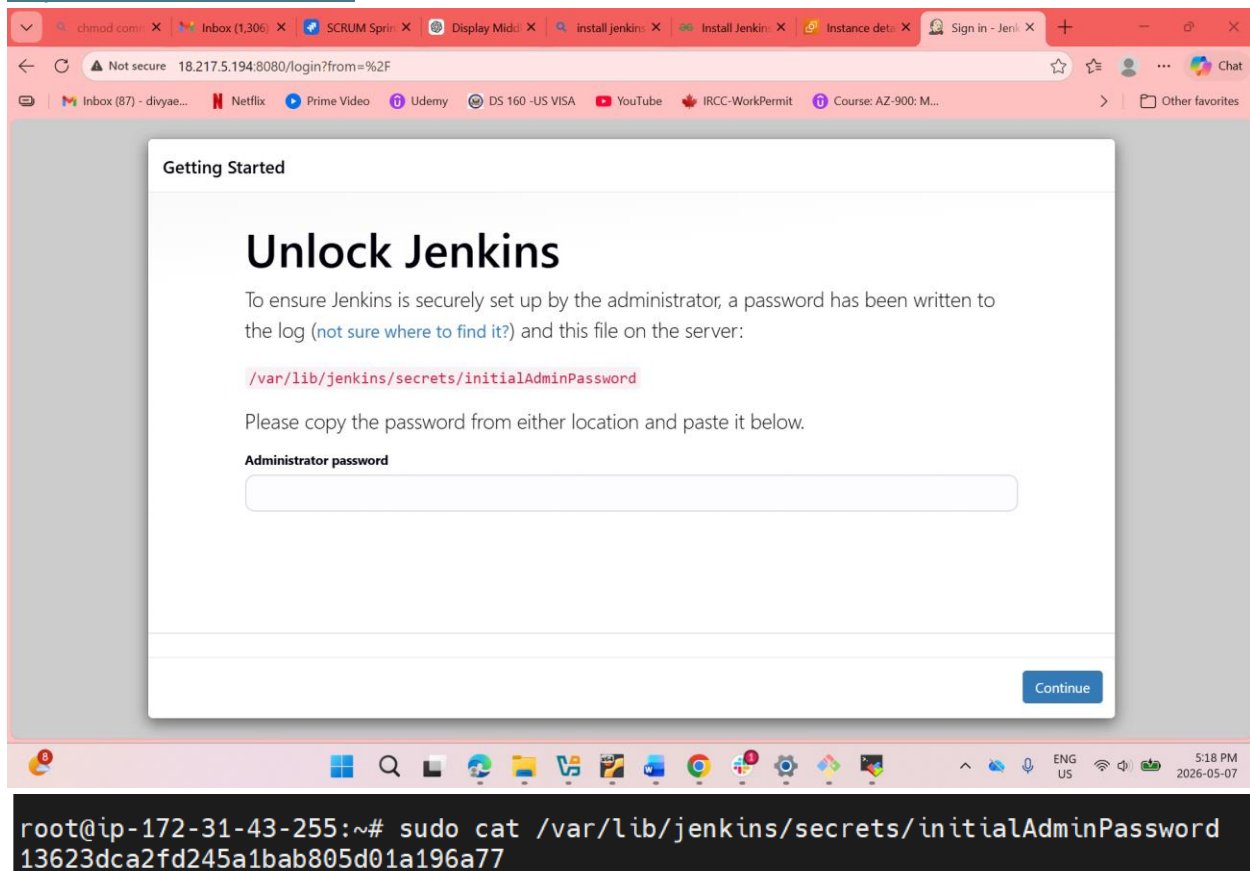
active (running) ---- Jenkins server is successfully running.

```
(Reading database ... 98958 files and directories currently installed.)
Preparing to unpack .../net-tools_2.10-2ubuntu1_amd64v3.deb ...
Unpacking net-tools (2.10-2ubuntu1) ...
Setting up net-tools (2.10-2ubuntu1) ...
Setting up jenkins (2.504.1) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/jenkins.service' -> '/usr/lib/systemd/system/jenkins.service'.
Processing triggers for man-db (2.13.1-1build1) ...
needrestart is being skipped since dpkg has failed
root@ip-172-31-43-255:~#
root@ip-172-31-43-255:~#
root@ip-172-31-43-255:~#
root@ip-172-31-43-255:~# sudo systemctl start jenkins
sudo systemctl enable jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
root@ip-172-31-43-255:~# sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-05-07 11:39:14 UTC; 32s ago
 Invocation: 077c8461e3374c2da6f1282d8c00e64b
    Main PID: 6345 (java)
      Tasks: 43 (limit: 627)
     Memory: 310.4M (peak: 343.7M)
        CPU: 23.218s
    CGroup: /system.slice/jenkins.service
            └─6345 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war

May 07 11:39:09 ip-172-31-43-255 jenkins[6345]: This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
May 07 11:39:09 ip-172-31-43-255 jenkins[6345]: *****
May 07 11:39:09 ip-172-31-43-255 jenkins[6345]: *****
May 07 11:39:09 ip-172-31-43-255 jenkins[6345]: *****
May 07 11:39:14 ip-172-31-43-255 jenkins[6345]: 2026-05-07 11:39:14.781+0000 [id=33] INFO jenkins.InitReactorRu
May 07 11:39:14 ip-172-31-43-255 jenkins[6345]: 2026-05-07 11:39:14.811+0000 [id=23] INFO hudson.lifecycle.Life
May 07 11:39:14 ip-172-31-43-255 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
May 07 11:39:15 ip-172-31-43-255 jenkins[6345]: 2026-05-07 11:39:15.089+0000 [id=49] INFO h.m.DownloadService$D
May 07 11:39:15 ip-172-31-43-255 jenkins[6345]: 2026-05-07 11:39:15.090+0000 [id=49] INFO hudson.util.Retrier#s
May 07 11:39:19 ip-172-31-43-255 jenkins[6345]: 2026-05-07 11:39:19.904+0000 [id=68] WARNING h.n.DiskSpaceMonit
Lines 1-21/21 (END)
```

URL used:

<http://18.217.5.194:8080>



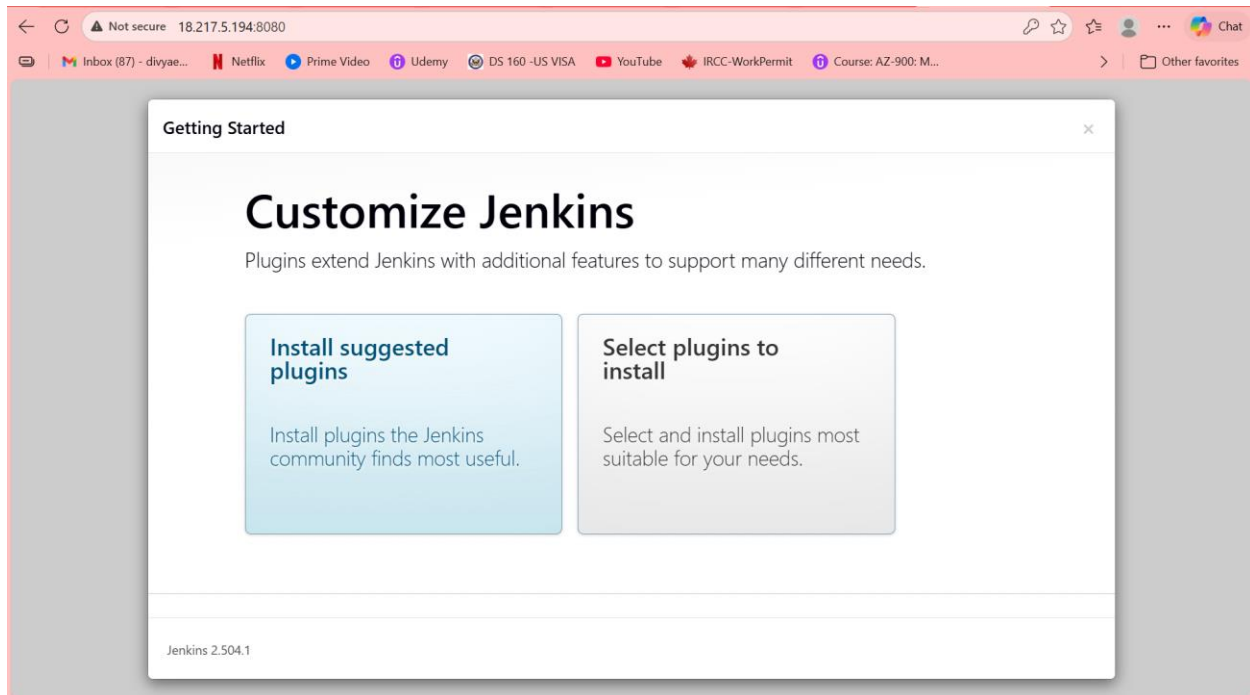
Retrieve Initial Admin Password

Command used:

```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

Purpose:

- Jenkins generates a temporary admin password during first startup.
- Needed for first-time login.



Tomcat Installation in Ubuntu VM

I have checked the tomcat in repository using command

```
sudo apt-cache search tomcat
```

confirmed that tomcat10 is available

```
ubuntu@ip-172-31-43-255:~$ sudo apt-cache search tomcat
centreon-plugins - Collection of Nagios plugins to monitor OS, services and network devices
css-validator-tomcat10 - W3C CSS Validator - tomcat10 configuration files
css-validator-tomcat11 - W3C CSS Validator - tomcat11 configuration files
libapache-mod-jk-doc - Documentation of libapache2-mod-jk package
libapache2-mod-jk - Apache 2 connector for the Tomcat Java servlet engine
libjnlpservlet-java - simple and convenient packaging format for JNLP applications
liblogback-java - flexible logging library for Java
libnetty-tcnative-java - Tomcat native fork for Netty
libnetty-tcnative-jni - Tomcat native fork for Netty (JNI library)
libspring-instrument-java - modular Java/J2EE application framework - Instrumentation
libtcnative-1 - Tomcat native library using the Apache Portable Runtime
libtomcat10-embed-java - Apache Tomcat 10 - Servlet and JSP engine -- embed libraries
libtomcat10-java - Apache Tomcat 10 - Servlet and JSP engine -- core libraries
libtomcat11-embed-java - Apache Tomcat 11 - Servlet and JSP engine -- embed libraries
libtomcat11-java - Apache Tomcat 11 - Servlet and JSP engine -- core libraries
libtomcat9-java - Apache Tomcat 9 - Servlet and JSP engine -- core libraries
monitoring-plugins-contrib - Plugins for nagios compatible monitoring systems
python3-ajpy - Python module to craft AJP requests
resource-agents-extra - Cluster Resource Agents
tomcat-jakartaee-migration - Apache Tomcat migration tool for Jakarta EE
tomcat10 - Apache Tomcat 10 - Servlet and JSP engine
tomcat10-admin - Apache Tomcat 10 - Servlet and JSP engine -- admin web applications
tomcat10-common - Apache Tomcat 10 - Servlet and JSP engine -- common files
tomcat10-docs - Apache Tomcat 10 - Servlet and JSP engine -- documentation
tomcat10-examples - Apache Tomcat 10 - Servlet and JSP engine -- example web applications
tomcat10-user - Apache Tomcat 10 - Servlet and JSP engine -- tools to create user instances
tomcat11 - Apache Tomcat 11 - Servlet and JSP engine
tomcat11-admin - Apache Tomcat 11 - Servlet and JSP engine -- admin web applications
tomcat11-common - Apache Tomcat 11 - Servlet and JSP engine -- common files
tomcat11-docs - Apache Tomcat 11 - Servlet and JSP engine -- documentation
tomcat11-examples - Apache Tomcat 11 - Servlet and JSP engine -- example web applications
tomcat11-user - Apache Tomcat 11 - Servlet and JSP engine -- tools to create user instances
yasat - simple stupid audit tool
ubuntu@ip-172-31-43-255:~$ sudo apt install tomcat9 tomcat9-admin
```

Install Tomcat 10

Using command ---- `sudo apt install tomcat10 tomcat10-admin -y`

Start Tomcat service:

```
sudo systemctl start tomcat10
```

Enable Tomcat service:

```
sudo systemctl enable tomcat10
```

Check Tomcat status:

```
sudo systemctl status tomcat10
```

If status shows: active (running)

then Tomcat is running successfully.


```

No services need to be restarted.

No containers need to be restarted.

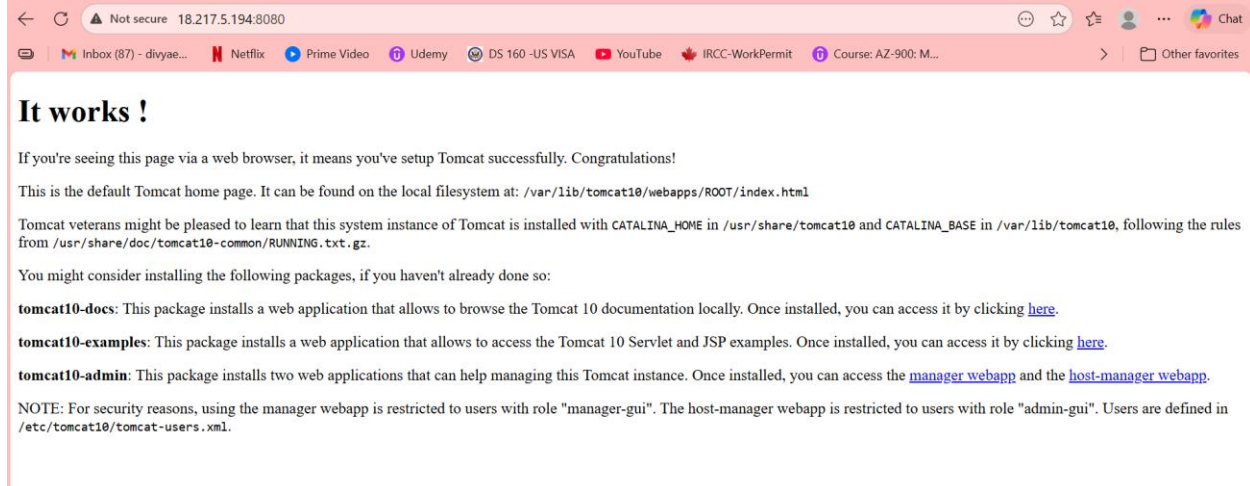
No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-43-255:~$
ubuntu@ip-172-31-43-255:~$
ubuntu@ip-172-31-43-255:~$
ubuntu@ip-172-31-43-255:~$
ubuntu@ip-172-31-43-255:~$ sudo systemctl start tomcat10
ubuntu@ip-172-31-43-255:~$ sudo systemctl enable tomcat10
ubuntu@ip-172-31-43-255:~$ sudo systemctl status tomcat10
● tomcat10.service - Apache Tomcat 10 Web Application Server
   Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-05-07 12:11:29 UTC; 30s ago
  Invocation: 10cd95c882e04e2a9558d0aa6155c861
     Docs: https://tomcat.apache.org/tomcat-10.0-doc/index.html
    Main PID: 8029 (java)
      Tasks: 18 (limit: 627)
    Memory: 94.2M (peak: 118.2M)
       CPU: 8.932s
    CGroup: /system.slice/tomcat10.service
            └─8029 /usr/lib/jvm/java-17-openjdk-amd64/bin/java -Djava.util.logging.config.file=/var/lib/tomcat10/conf/logging
May 07 12:11:34 ip-172-31-43-255 tomcat10[8029]: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug lo
May 07 12:11:34 ip-172-31-43-255 tomcat10[8029]: Deployment of deployment descriptor [/etc/tomcat10/Catalina/localhost/host-m
May 07 12:11:34 ip-172-31-43-255 tomcat10[8029]: Deploying deployment descriptor [/etc/tomcat10/Catalina/localhost/manager.xml
May 07 12:11:34 ip-172-31-43-255 tomcat10[8029]: The path attribute with value [/manager] in deployment descriptor [/etc/tomc
May 07 12:11:35 ip-172-31-43-255 tomcat10[8029]: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug lo
May 07 12:11:35 ip-172-31-43-255 tomcat10[8029]: Deployment of deployment descriptor [/etc/tomcat10/Catalina/localhost/manager
May 07 12:11:35 ip-172-31-43-255 tomcat10[8029]: Deploying web application directory [/var/lib/tomcat10/webapps/ROOT]
May 07 12:11:36 ip-172-31-43-255 tomcat10[8029]: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug lo
May 07 12:11:36 ip-172-31-43-255 tomcat10[8029]: Deployment of web application directory [/var/lib/tomcat10/webapps/ROOT] has
May 07 12:11:36 ip-172-31-43-255 tomcat10[8029]: Server startup in [4808] milliseconds
lines 1-22/22 (END)

```

access Tomcat in browser:

http:// 18.217.5.194:8080



Jenkins is already using port 8080

Tomcat also uses port 8080 by default . So Tomcat may failed to start due to port conflict

To stop Jenkins: `sudo systemctl stop Jenkins`

Then restart Tomcat: `sudo systemctl restart tomcat10`

Setup Docker Desktop on your local windows.

